UNIVERSIDAD SAN FRANCISCO DE QUITO
COLEGIO: CIENCIAS E INGENIERÍAS
Semestre: 202520 - Segundo Semestre 2025/2026

Phase 3 Final Project

Simulated Bitcoin Exchange

How Phase 3 will be graded

This phase will be also graded by running your published Docker images directly from Docker Hub. Your code will not be rebuilt locally during grading. All published images must run on both ARM and AMD CPUs.

You must still upload the full source code to D2L. Include a README.txt file with the submission. The README.txt must list all team members exactly as they appear in D2L, describe each person's contribution, and provide complete instructions for running this phase of the project.

Each phase builds on the previous one. The project consists of three phases, and the third phase is the final project.

1. Additional Functionality

This stage adds a client history view.
Each client account must have a chronological history of its BTC-related events.
The history must include:

- BTC buy events
- BTC sell events
- BTC transfer events made by the client

There must be a UI table that allows viewing the history for each client.

The table must contain the following columns:

a) A unique account identifier.
b) The client name.
c) The timestamp of the event.
d) The event type:

- Buy BTC
- Sell BTC
- Transfer BTC

e) The event details, depending on the event type:

- for **Buy BTC**: BTC quantity and price per BTC
- for **Sell BTC**: BTC quantity and price per BTC
- for **Transfer BTC**: BTC amount withdrawn from the exchange
    - In this simulation all the BTC held by clients was either transferred to the Exchange in the case of miners or obtained through the Exchange by the other clients that started with a dollar amount in the Exchange.

Clarification:
In this simulation, all BTC held by clients comes either from BTC transferred into the exchange by miners or from BTC purchased through the exchange by other clients who began with a USD balance inside the exchange.

2. Presentation

Stage 3 should also include a presentation that explains how the complete exchange works and how Stages 1, 2, and 3 fit together into one coherent system. The presentation should connect the mining and exchange-entry logic from Stage 1, the trading and order-handling logic from Stage 2, and the history and final-system explanation from Stage 3.